

Unit 3

CHEMICAL BONDING

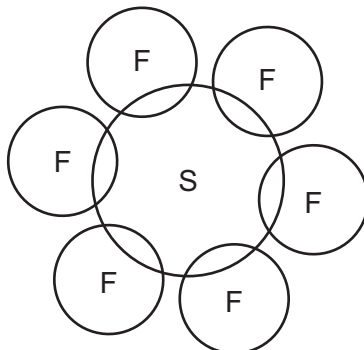
57 items · 145 marks

Syllabus sections in this unit

- 3.1 Electronegativity and bonding
- 3.2 Ionic bonding
- 3.3 Metallic bonding
- 3.4 Covalent bonding and coordinate (dative covalent) bonding
- 3.5 Shapes of molecules
- 3.6 Intermolecular forces, electronegativity and bond properties
- 3.7 Dot-and-cross diagrams

Unit 3: Chemical bonding**1. M/J 26/P21/Q1(a)**

- 1 (a) (i) Complete the dot-and-cross diagram in Figure 1.1 to show the arrangement of outer shell electrons in a molecule of SF_6 .

**Figure 1.1**

- (ii) Name the shape of a molecule of SF_6 .

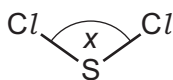
[1]

[1]

2. M/J 26/P21/Q1(c)

- (c) A molecule of SCl_2 is polar.

- (i) Predict bond angle x in Figure 1.2 using VSEPR theory. Explain your answer.

**Figure 1.2**

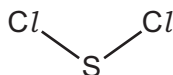
bond angle $x =$

°

explanation

[3]

- (ii) Show on Figure 1.3 all the dipoles present in the molecule of SCl_2 .

**Figure 1.3**

[1]

3. M/J 26/P21/Q1(d)

(d) SCl_2 and H_2O both contain instantaneous dipole–induced dipole attractions and permanent dipole attractions.

(i) Complete Table 1.2.

Table 1.2

	SCl_2	H_2O
total number of electrons per molecule		10

[1]

(ii) Identify which molecule, SCl_2 or H_2O , has the stronger instantaneous dipole–induced dipole attractions. Explain your answer.

[1]

(iii) H_2O has a greater dipole moment than SCl_2 . Explain why.

[1]

4. M/J 26/P22/Q1(b)

Context

1 Oxygen reacts with many elements.

(b) Lithium reacts with oxygen to produce lithium oxide, Li_2O .

Li_2O is a white solid with a melting point over 1400°C .

Draw a dot-and-cross diagram to show the arrangement of **all** the electrons in Li_2O .

[2]

5. M/J 26/P24/Q1(c)

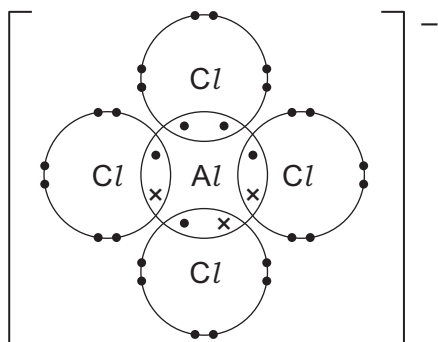
Context

- 1** Aluminium is a Group 13 element that forms a number of compounds.

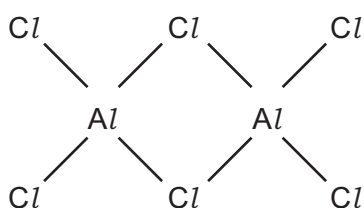
Table 1.1

atom or ion	number of protons	number of neutrons
^{27}Al		
$^{28}\text{Al}^{3+}$		

Context: Fig. 1.3

**Figure 1.3**

- (c) AlCl_3 dimerises to form Al_2Cl_6 . Al_2Cl_6 is shown in Figure 1.2.

**Figure 1.2**

Describe how coordinate (dative covalent) bonds form when two AlCl_3 molecules combine to form Al_2Cl_6 .

[1]

6. M/J 26/P24/Q1(d)

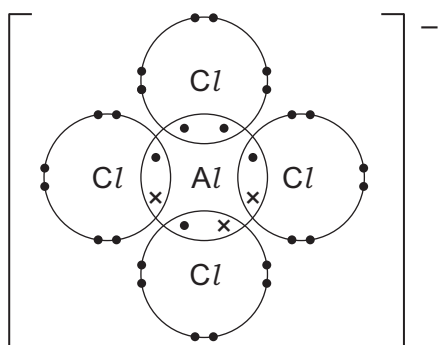
Context

- 1** Aluminium is a Group 13 element that forms a number of compounds.

Table 1.1

atom or ion	number of protons	number of neutrons
^{27}Al		
$^{28}\text{Al}^{3+}$		

- (d)** Aluminium forms the AlCl_4^- ion. The dot-and-cross diagram for AlCl_4^- in Figure 1.3 shows the arrangement of outer electrons.

**Figure 1.3**

- (i)** Use Figure 1.3 and VSEPR theory to name the shape of the AlCl_4^- ion and to predict the $\text{Cl}-\text{Al}-\text{Cl}$ bond angle.

name of shape

bond angle

[2]

- (ii)** Draw the three-dimensional shape of the AlCl_4^- ion.

[1]

7. M/J 26/P24/Q2(a)(i)

- 2 (a) (i)** Nitrogen, N_2 , contains sigma (σ) and pi (π) bonds.

Describe how a pi (π) bond is formed.

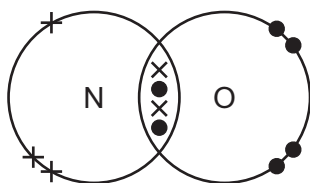
[2]

8. MAR 26/P22/Q2(a)(ii)

Context

2 Nitrogen monoxide, NO, is an atmospheric pollutant.

Context: part (a) (introduction)

(a) Figure 2.1 shows the dot-and-cross diagram of NO.**Figure 2.1****(ii)** In NO, the N and O atoms are bonded by a sigma (σ) and a pi (π) bond.

Complete Figure 2.2 to show the pi bond in NO that is formed from orbital overlap.

**Figure 2.2**

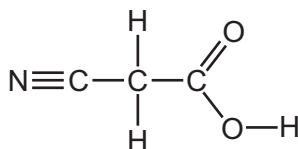
[1]

9. MAR 26/P22/Q4(b)

Context

4 Cyanoethanoic acid, $\text{CH}_2(\text{CN})\text{COOH}$, is a weak organic acid.**(b)** $\text{CH}_2(\text{CN})\text{COOH}$ is very soluble in water because it can form hydrogen bonds with water.Complete Figure 4.2 to show **one** hydrogen bond between a molecule of $\text{CH}_2(\text{CN})\text{COOH}$ and a molecule of H_2O .

Label the hydrogen bond clearly, and show any relevant partial charges and lone pairs of electrons.

**Figure 4.2**

[3]

10. O/N 25/P22/Q3(d)(ii)

Context

- 3** The alkanes are a homologous series of organic molecules. Alkanes are generally unreactive and are commonly used as fuels.

Context: part (d) (introduction)

- (d)** Hexane, C_6H_{14} , has four structural isomers.

Fig. 3.1 shows hexane and two of its structural isomers.

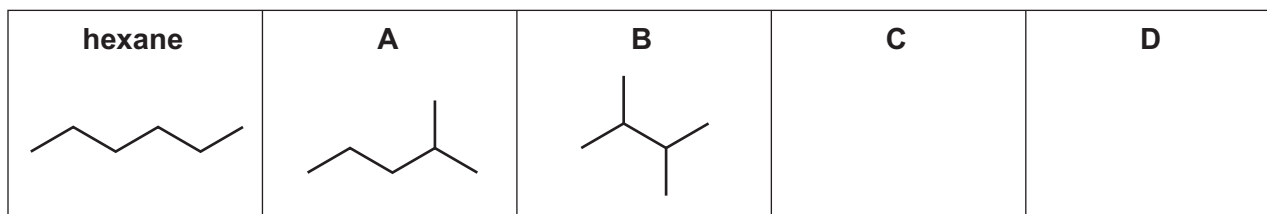


Fig. 3.1

- (ii)** **A**, **B** and hexane have different boiling points.

Arrange **A**, **B** and hexane in order of increasing boiling point.

Explain your answer.

lowest < < highest
[3]

11. O/N 25/P24/Q1(b)(v)

Context

- 1** The Group 17 elements are oxidising agents.

Context: part (b) (introduction)

- (b)** Cl_2 and Br_2 can each react with NH_3 to give N_2 and a hydrogen halide, HX .



The relative bond strengths of $X-X$ and $H-X$ determine the difference in enthalpy change of the two reactions.

- (v)** I_2 reacts with NH_3 to form NI_3 .

Predict the shape of a molecule of NI_3 . Explain your answer.

shape

explanation

[2]

12. M/J 25/P21/Q1(a)(ii)

Context: part (a) (introduction)

- 1** **(a)** Solid sodium conducts electricity. Sodium oxide conducts electricity when molten but not when solid.
- (ii)** Explain how solid sodium conducts electricity.

[1]

13. M/J 25/P23/Q4(b)

Context

- 4** The reaction between $\text{Na}_2\text{S}_2\text{O}_3(\text{aq})$ and $\text{HCl}(\text{aq})$ is monitored at constant temperature.

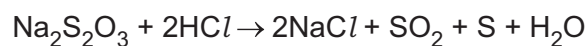
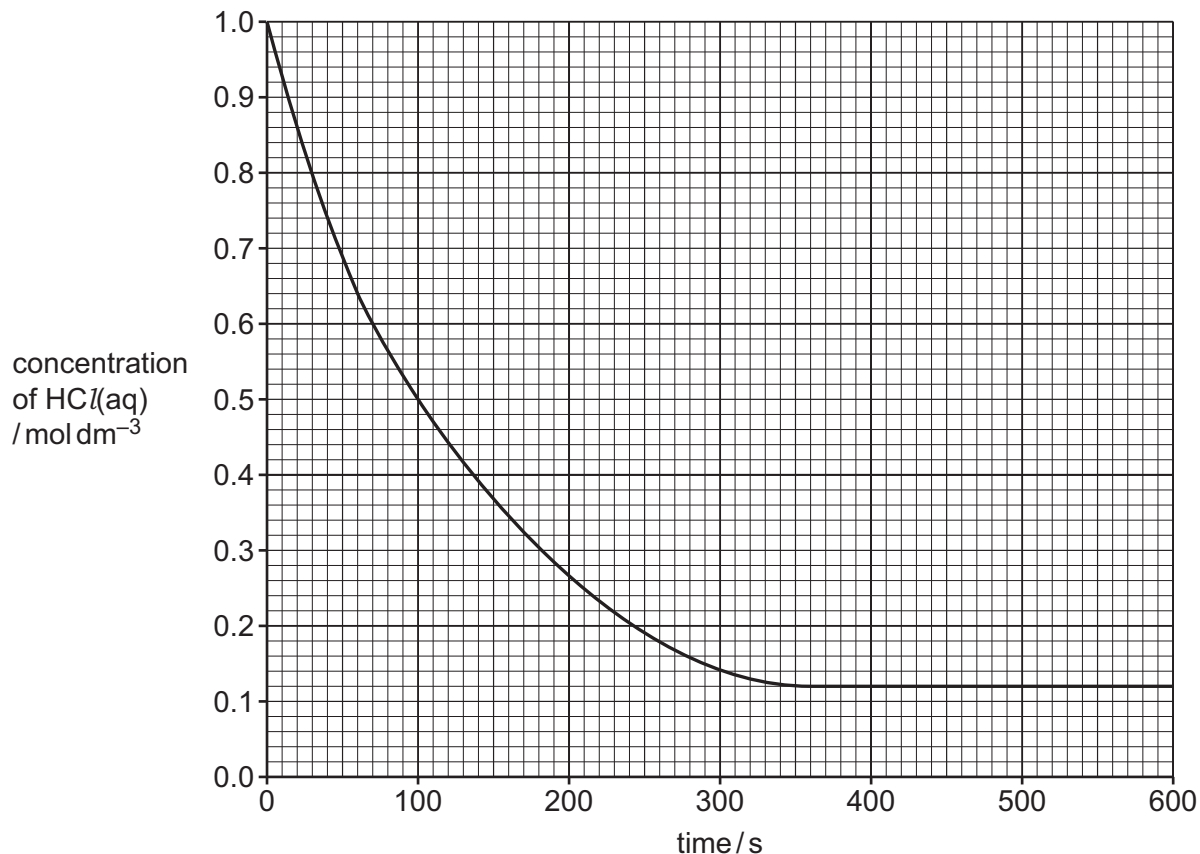
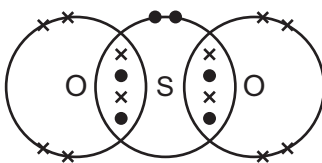


Fig. 4.1 shows how the concentration of $\text{HCl}(\text{aq})$ varies with time.

**Fig. 4.1**

- (b)** Fig. 4.2 shows a possible arrangement of outer-shell electrons in one SO_2 molecule.

**Fig. 4.2**

- (i)** Use Fig. 4.2 to predict the shape and bond angle of a molecule of SO_2 .

shape

bond angle

°

[2]

- (ii)** Use Table 4.1 to predict the strength of the dipole moment of SO_2 , if any, compared to that of H_2O . Explain your answer.

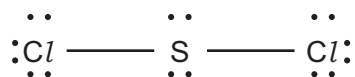
Table 4.1

	H	O	S
electronegativity	2.1	3.5	2.6

[2]

14. M/J 25/P24/Q2(a)

- 2 (a) Fig. 2.1 shows the covalent bonds and lone pairs of electrons in a molecule of SCl_2 .

**Fig. 2.1**

- (i) Use VSEPR theory to predict the shape of a molecule of SCl_2 .

[1]

- (ii) Predict the bond angle in a molecule of SCl_2 .

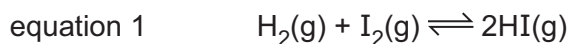
°

[1]

15. M/J 25/P24/Q3(d)(i)

Context

- 3 $\text{H}_2(\text{g})$ and $\text{I}_2(\text{g})$ react to form $\text{HI}(\text{g})$ in a reversible reaction, as shown in equation 1.



In three separate experiments, a student combines different amounts of two or more of the gases from equation 1. In each experiment, the gases are left to reach equilibrium at a given temperature.

Context: part (d) (introduction)

- (d) The value of K_c for the dissociation of HCl at 298 K is 5.50×10^{-34} .



- (i) Define covalent bond.

[1]

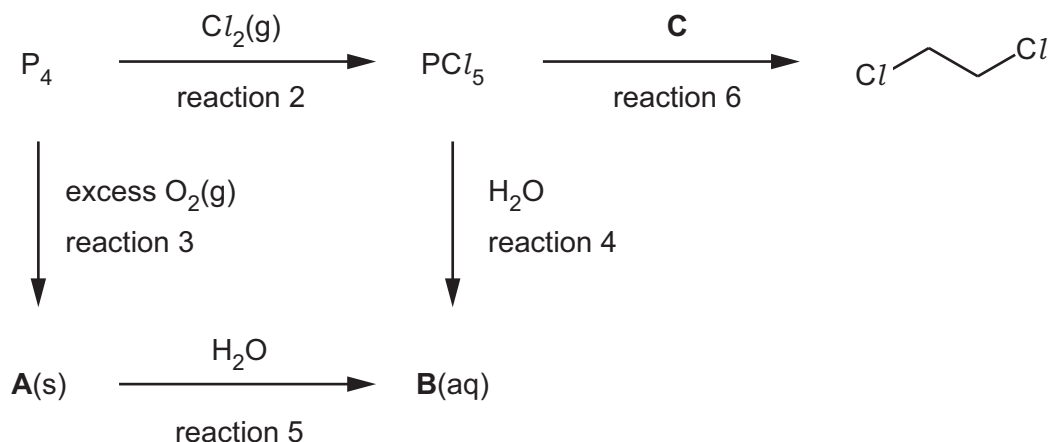
16. MAR 25/P22/Q1(b)(ii)

Context

1 Phosphorus and chlorine are elements in Period 3 of the Periodic Table.

Context: part (b) (introduction)

(b) Fig. 1.1 shows a reaction scheme involving P_4 and Cl_2 .

**Fig. 1.1**

(ii) Predict the shape of a molecule of PCl_5 .

[1]

17. MAR 25/P22/Q2(a)

2 **(a)** The Group 2 elements show metallic bonding.

Define metallic bonding.

[2]

18. O/N 24/P21/Q3(a)(iii)

Context

3 The Group 14 elements show a change from non-metallic to metallic character down the group.

Context: part (a) (introduction)

(a) Table 3.1 shows some properties of two Group 14 elements, C and Sn, in their standard states. The table is incomplete.

Table 3.1

	C (graphite)	Sn
state and appearance in standard state	grey shiny solid	silvery solid
electrical conductivity		good
type of bonding		metallic
type of structure	giant	

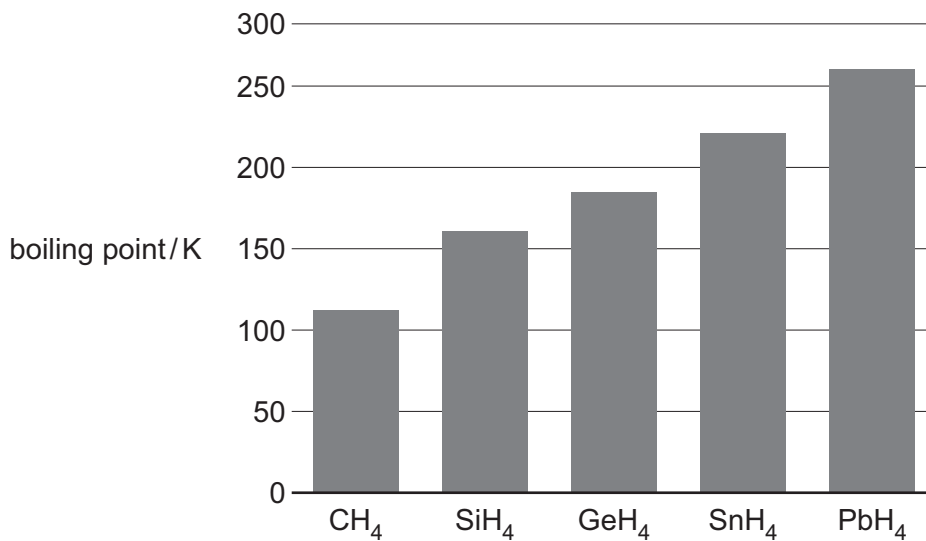
(iii) Explain why Sn has good electrical conductivity.

[1]

19. O/N 24/P21/Q3(d)

Context

- 3** The Group 14 elements show a change from non-metallic to metallic character down the group.
- (d)** Fig. 3.2 shows the boiling points of the simplest hydrides of the Group 14 elements, C to Pb.

**Fig. 3.2**

- (i)** Explain the trend in the boiling points of the Group 14 hydrides shown in Fig. 3.2.

[2]

- (ii)** Deduce the shape of a molecule of SiH₄.

[1]

20. O/N 24/P22/Q2(a)(ii)

Context

- 2** Oxygen is a Group 16 element.

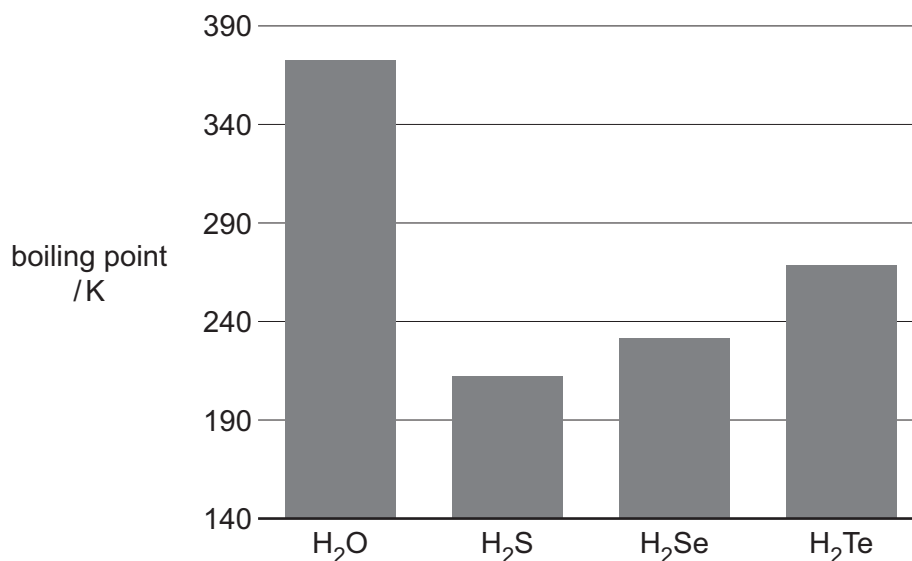
- (ii)** Draw a dot-and-cross diagram to show the species present in Al₂O₃.

Draw outer electrons only.

[1]

21. O/N 24/P22/Q2(c)

Context

2 Oxygen is a Group 16 element.**(c)** Fig. 2.1 shows the boiling points of H_2O and other Group 16 hydrides.**Fig. 2.1****(i)** Explain the trend in the boiling points of the Group 16 hydrides H_2S to H_2Te .

[2]

(ii) Explain why the boiling point of H_2O is much higher than that of H_2S .

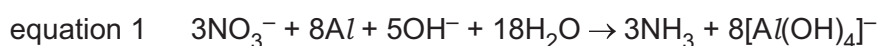
[1]

22. O/N 24/P22/Q3(b)(iv)

Context

3 Nitrogen and phosphorus are elements in Group 15 of the Periodic Table.

Context: part (b) (introduction)

(b) A common test for nitrates is the reaction with NaOH and Al . Equation 1 shows the reaction.**(iv)** Suggest the shape of the $[\text{Al}(\text{OH})_4]^-$ ion.

[1]

23. O/N 24/P22/Q3(f)(i)

Context

3 Nitrogen and phosphorus are elements in Group 15 of the Periodic Table.

Context: part (f) (introduction)

(f) At very high temperatures, phosphorus can form P_2 molecules. P_2 contains a triple bond, $\text{P}\equiv\text{P}$.**(i)** Describe the formation of the $\text{P}\equiv\text{P}$ bond in terms of orbital overlap.

[2]

24. O/N 24/P23/Q3(a)(iii)

Context

3 The Group 14 elements show a change from non-metallic to metallic character down the group.

Context: part (a) (introduction)

(a) Table 3.1 shows some properties of two Group 14 elements, C and Sn, in their standard states. The table is incomplete.

Table 3.1

	C (graphite)	Sn
state and appearance in standard state	grey shiny solid	silvery solid
electrical conductivity		good
type of bonding		metallic
type of structure	giant	

(iii) Explain why Sn has good electrical conductivity.

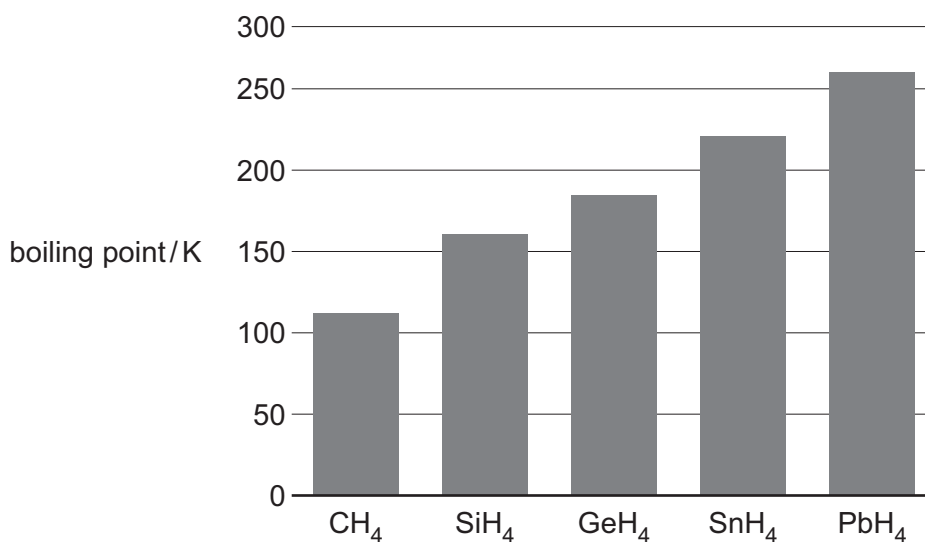
[1]

25. O/N 24/P23/Q3(d)

Context

3 The Group 14 elements show a change from non-metallic to metallic character down the group.

(d) Fig. 3.2 shows the boiling points of the simplest hydrides of the Group 14 elements, C to Pb.

**Fig. 3.2**

(i) Explain the trend in the boiling points of the Group 14 hydrides shown in Fig. 3.2.

[2]

(ii) Deduce the shape of a molecule of SiH₄.

[1]

26. M/J 24/P21/Q2(b)

(b) A molecule of SCl_2 contains two S–Cl covalent bonds.

- (i) Complete the dot-and-cross diagram in Fig. 2.1 to show the arrangement of the outer electrons in a molecule of SCl_2 .

Use $\times$ to show electrons from the chlorine atoms.

Use $\bullet$ to show electrons from the sulfur atom.

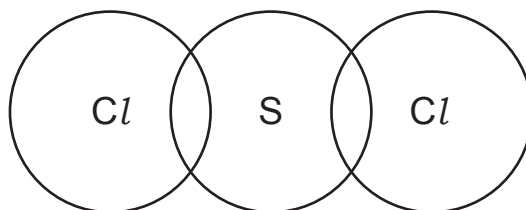


Fig. 2.1

[2]

- (ii) Predict the shape of, and bond angle in, a molecule of SCl_2 by using VSEPR theory.

shape

bond angle

[2]

27. M/J 24/P21/Q5(a)

Context

- 5** Hydrocarbon molecules contain covalent bonds.

(a) Define covalent bond.

[1]

28. M/J 24/P22/Q1(c)

(c) Draw a labelled diagram to show the structure and bonding in sodium.

[1]

29. M/J 24/P22/Q3(d)

(d) $\text{SCN}^-(\text{aq})$ is colourless.

Complete the dot-and-cross diagram in Fig. 3.1 to show the arrangement of outer electrons in an SCN^- ion.

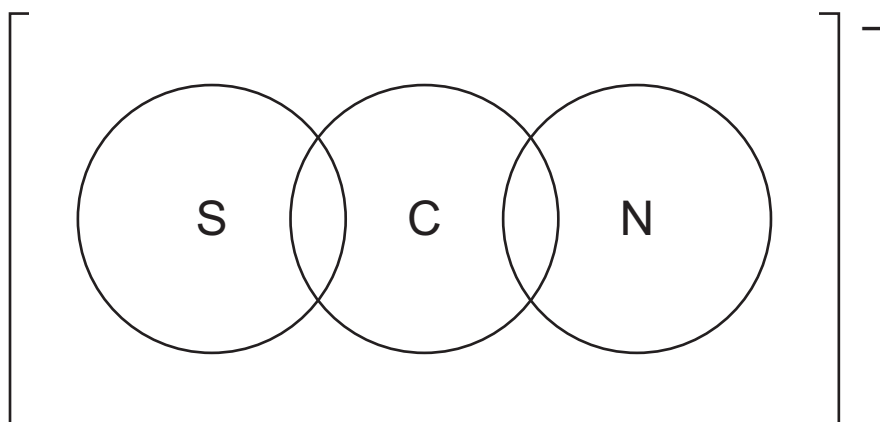


Fig. 3.1

[2]

30. M/J 24/P23/Q1(a)(ii)

(ii) Covalent bonds can be σ bonds or π bonds.

Complete Table 1.1 to show the number of σ and π bonds in a molecule of N_2 and to describe how the orbitals overlap to form σ and π bonds.

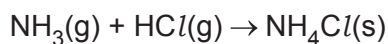
Table 1.1

	σ bond	π bond
number of bonds in N_2		
how the orbitals overlap		

[4]

31. M/J 24/P23/Q4(a)

4 (a) $\text{NH}_3(\text{g})$ reacts with $\text{HCl}(\text{g})$ to produce $\text{NH}_4\text{Cl}(\text{s})$, as shown.



Draw a diagram to show the ionic, covalent and coordinate bonding present in a formula unit of NH_4Cl .

[2]

32. MAR 24/P22/Q1(a)

Context

- 1** Bismuth is an element in Group 15 of the Periodic Table.

(a) Bismuth has metallic bonding.

Draw a labelled diagram to show the metallic bonding in bismuth.

[1]

33. MAR 24/P22/Q2(c)(i)

Context

- 2** Chlorine, Cl_2 , reacts with many elements and compounds to form chlorides.

Table 2.1 shows information about some chlorides of Period 3 elements.

Table 2.1

	Na	Mg	Si
formula of chloride			
structure of chloride	giant		
bonding of chloride			covalent
pH of solution formed on addition of chloride to water		6.2	

Context: part (c) (introduction)

- (c) An excess of Cl_2 reacts with phosphorus to form PCl_5 .

(i) PCl_5 is a simple molecule in the gas phase.

It also exists in a solid form as two ions, PCl_4^+ and PCl_6^- .

Complete Table 2.2 to identify the shapes of each of these species.

Table 2.2

species	PCl_5	PCl_4^+	PCl_6^-
shape		tetrahedral	

[2]

34. MAR 24/P22/Q3(d)

Context

3 Nitrogen, N_2 , is generally an unreactive molecule but it does react under certain conditions.

(d) The N_2F_2 molecule has a double covalent bond between its nitrogen atoms. This consists of a σ and a π bond.

(i) Complete Fig. 3.2 to show the dot-and-cross diagram for N_2F_2 .

Show outer electrons only.

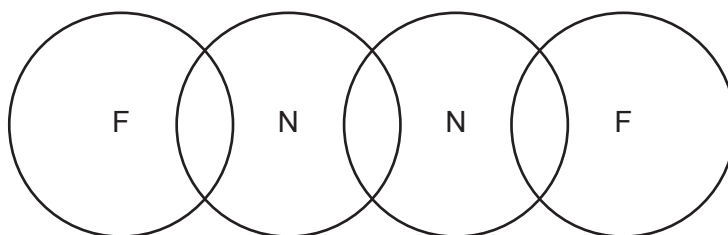


Fig. 3.2

[2]

(ii) Deduce the hybridisation of the N atoms in N_2F_2 .

[1]

(iii) Draw a diagram of the π bond between the N atoms in N_2F_2 and describe how it forms.

[2]

35. O/N 23/P21/Q1(d)(iii)

Context

- 1** The elements phosphorus, sulfur and chlorine are in Period 3 of the Periodic Table.

Table 1.1 shows some properties of the elements P to Cl.

The first ionisation energy of S is **not** shown.

Table 1.1

property	P	S	Cl
number of electrons in 3p subshell			
total number of unpaired electrons			
first ionisation energy /kJ mol ⁻¹	1060		1260
formula of most common anion	P ³⁻	S ²⁻	Cl ⁻

Context: part (d) (introduction)

- (d)** POC_l₃ shows similar chemical properties to PCl₅.

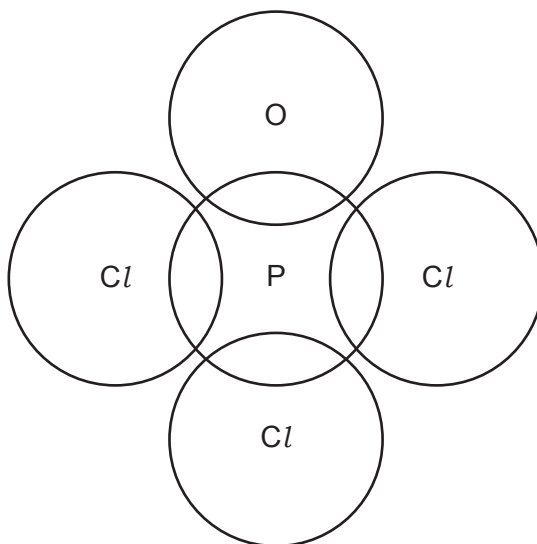
POC_l₃ has a melting point of 1°C and a boiling point of 106°C.

POC_l₃ reacts vigorously with water, forming misty fumes and an acidic solution.

- (iii)** POC_l₃ contains a double covalent bond between P and O.

Complete the dot-and-cross diagram, in Fig. 1.1, to show the bonding in POC_l₃.

Show outer shell electrons only.

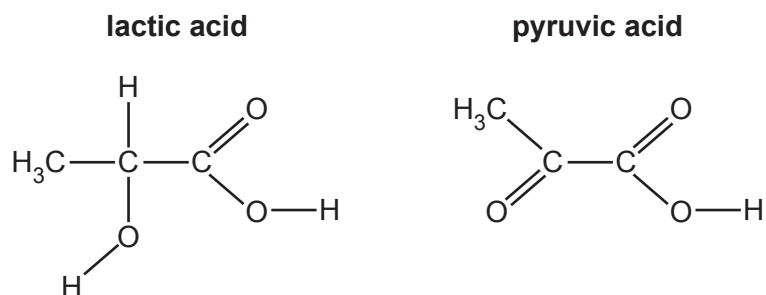
**Fig. 1.1**

[2]

36. O/N 23/P22/Q4(a)(iii)

Context

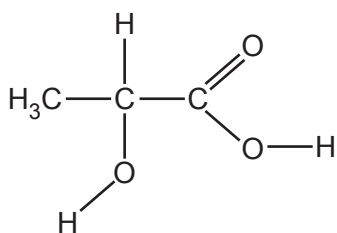
- 4** Lactic acid, $\text{CH}_3\text{CH}(\text{OH})\text{COOH}$, and pyruvic acid, CH_3COCOOH , both contain two functional groups.

**Fig. 4.1**

- (iii)** Lactic acid forms hydrogen bonds with water.

Complete Fig. 4.2 to show the formation of a hydrogen bond between one molecule of lactic acid and one molecule of water.

Label the hydrogen bond. Show any relevant dipoles and lone pairs of electrons.

**Fig. 4.2**

[3]

37. O/N 23/P23/Q1(d)(iii)

Context

- 1** The elements phosphorus, sulfur and chlorine are in Period 3 of the Periodic Table.

Table 1.1 shows some properties of the elements P to Cl.

The first ionisation energy of S is **not** shown.

Table 1.1

property	P	S	Cl
number of electrons in 3p subshell			
total number of unpaired electrons			
first ionisation energy /kJ mol ⁻¹	1060		1260
formula of most common anion	P ³⁻	S ²⁻	Cl ⁻

Context: part (d) (introduction)

- (d)** POC_l₃ shows similar chemical properties to PCl₅.

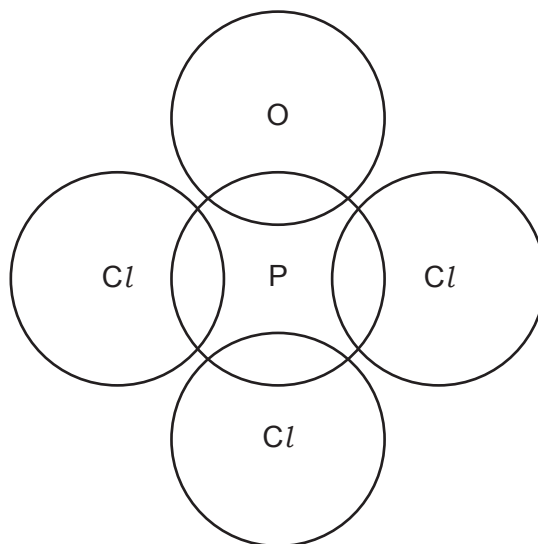
POC_l₃ has a melting point of 1°C and a boiling point of 106°C.

POC_l₃ reacts vigorously with water, forming misty fumes and an acidic solution.

- (iii)** POC_l₃ contains a double covalent bond between P and O.

Complete the dot-and-cross diagram, in Fig. 1.1, to show the bonding in POC_l₃.

Show outer shell electrons only.

**Fig. 1.1**

38. M/J 23/P21/Q1(e)

Context

- 1** Tellurium is an element in Group 16. The most common isotope of tellurium is ^{130}Te . Its electronic configuration is $[\text{Kr}] 4d^{10} 5s^2 5p^4$.
- (e)** Te reacts with F_2 at 150°C to form TeF_x . Molecules of TeF_x are octahedral with bond angles of 90° .

Explain why TeF_x is octahedral with bond angles of 90° .

[2]

39. M/J 23/P22/Q1(b)

Context

- 1** The melting points of some solids are shown in Table 1.1.

Table 1.1

solid	melting point / K
magnesium	923
phosphorus	317
sodium chloride	1074
sulfur	392

- (b) (i) Define electronegativity.

[1]

- (ii) Explain why electronegativity increases across a period.

[2]

- (iii) Name the strongest intermolecular force that exists between $\text{NH}_3(\text{l})$ molecules.

[1]

- (iv) Draw a diagram to show the formation of the strongest intermolecular force between **two** molecules of $\text{NH}_3(\text{l})$.

Include any relevant lone pairs of electrons and dipoles.

[2]

- (v) The melting points of ice and ammonia are shown in Table 1.2.

Table 1.2

solid	melting point / K
ice	273
ammonia	195

Suggest **two** reasons for the difference in the melting points of ice and ammonia.

[2]

40. M/J 23/P23/Q1(a)(ii)

Context

- 1** Copper is used in electrical equipment. It has a melting point of 1085°C .

- (ii) Draw a labelled diagram to show the bonding present in copper.

[1]

41. MAR 23/P22/Q1(a)

Context

- 1 The Pauling electronegativity values of elements can be used to predict the chemical properties of compounds.

Use the information in Table 1.1 to answer the following questions.

Table 1.1

element	H	Li	C	O	S
Pauling electronegativity value	2.1	1.0	2.5	3.5	2.6
first ionisation energy/kJ mol ⁻¹	1310	519	1090	1310	1000
second ionisation energy/kJ mol ⁻¹	—	7300	2350	3390	2260

- (a) (i) Define electronegativity.

[1]

- (ii) O and S are in Group 16.

Explain the difference in the Pauling electronegativity values of O and S.

[2]

42. MAR 23/P22/Q1(b)

Context

- 1 The Pauling electronegativity values of elements can be used to predict the chemical properties of compounds.

Use the information in Table 1.1 to answer the following questions.

Table 1.1

element	H	Li	C	O	S
Pauling electronegativity value	2.1	1.0	2.5	3.5	2.6
first ionisation energy/kJ mol ⁻¹	1310	519	1090	1310	1000
second ionisation energy/kJ mol ⁻¹	—	7300	2350	3390	2260

- (b) (i) LiH is an ionic compound.

Draw a dot-and-cross diagram of LiH.

Include **all** electrons.

[2]

- (ii) Suggest the shape of a molecule of H₂S.

[1]

43. MAR 23/P22/Q1(e)(i)

Context

- 1** The Pauling electronegativity values of elements can be used to predict the chemical properties of compounds.

Use the information in Table 1.1 to answer the following questions.

Table 1.1

element	H	Li	C	O	S
Pauling electronegativity value	2.1	1.0	2.5	3.5	2.6
first ionisation energy / kJ mol ⁻¹	1310	519	1090	1310	1000
second ionisation energy / kJ mol ⁻¹	—	7300	2350	3390	2260

- (e) (i)** Complete Table 1.2 by placing a tick (✓) to show which of the compounds have molecules with an overall dipole moment.

Table 1.2

compound	O=C=O	O=S=O	S=C=S	S=C=O
overall dipole moment				

[2]

44. MAR 23/P22/Q2(a)

Context

- 2** The Group 2 elements Mg to Ba are all silvery-white reactive metals.

- (a) (i)** Draw a labelled diagram to show the bonding and structure of the Group 2 metals at room temperature.

[2]

- (ii)** Explain why Mg has a higher electrical conductivity than Na.

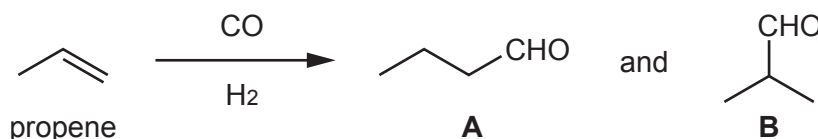
[1]

45. MAR 23/P22/Q3(d)(iv)

Context

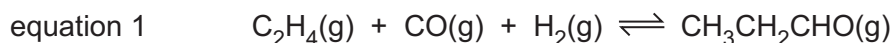
- 3** Alkenes undergo an addition reaction with a 1:1 mixture of CO and H₂ to form aldehydes.

Fig. 3.1 shows the reaction of propene with a 1:1 mixture of CO and H₂.

**Fig. 3.1**

Context: part (d) (introduction)

- (d)** The reaction of ethene, C₂H₄, with a 1:1 mixture of CO and H₂ is shown in equation 1.



At atmospheric pressure a cobalt-based catalyst is used in this reaction.

- (iv)** The reaction mixture is cooled to collect CH₃CH₂CHO as a liquid.

Identify all types of van der Waals' forces that are present between molecules of CH₃CH₂CHO.

[1]

46. O/N 22/P21/Q2(a)

Context

- 2** Magnesium shows reactions typical of a Group 2 metal.

- (a)** Draw a labelled diagram to show the bonding in magnesium metal.

[2]

47. O/N 22/P21/Q3(d)

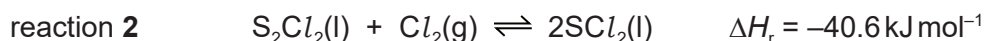
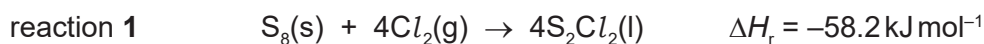
also Unit 4 (2 marks), Unit 5 (2 marks), Unit 13 (2 marks), Unit 7 (1 mark); this unit 4 marks

Context

- 3** Some of the common chlorides of Period 3 elements are shown in the list.



- (d)** Sulfur, S₈, reacts with chlorine to form several different chlorides. The most common are S₂Cl₂ and SCl₂. SCl₂ forms when sulfur reacts with an excess of chlorine.



- (i)** SCl₂ is a cherry-red liquid that reacts vigorously with water to form an acidic solution.

Use this information to deduce the bonding and structure shown by SCl₂.
Explain your answer.

[2]

- (ii) Calculate the enthalpy change of formation, ΔH_f , of $\text{SCl}_2(\text{l})$. You may find it useful to use Hess's Law to construct an energy cycle.

enthalpy change of formation of $\text{SCl}_2(\text{l})$, $\Delta H_f =$ kJ mol^{-1}
[2]

- (iii) State the effect of a decrease in pressure on the position of equilibrium in reaction 2. Explain your answer.

[1]

Fig. 3.1 shows the two structural isomers of S_2Cl_2 .

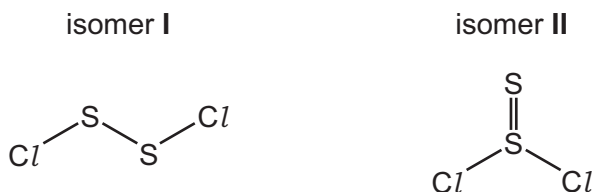


Fig. 3.1

- (iv) Define the term structural isomer.

[2]

- (v) Suggest a value for the Cl-S-S bond angle in isomer I. Explain your answer.

bond angle = °

explanation

[2]

- (vi) Draw a dot-and-cross diagram to show the bonding in isomer II. Show outer shell electrons only.

[2]

48. O/N 22/P22/Q1(a)

Context

- 1 Species such as NH_4^+ , CO_3^{2-} and PO_4^{3-} are examples of molecular ions.

- (a) Ionic and covalent bonds both involve an electrostatic attraction between different species.

Identify the species that are electrostatically attracted to one another in:

- an ionic bond
- a covalent bond.

[2]

49. O/N 22/P22/Q1(c)(iii)

Context

1 Species such as NH_4^+ , CO_3^{2-} and PO_4^{3-} are examples of molecular ions.

Context: part (c) (introduction)

(c) NH_4^+ is a Brønsted–Lowry acid.

(iii) The nitrogen atom in NH_4^+ is sp^3 hybridised. sp^3 orbitals form from the mixing of one 2s and three 2p orbitals.

Sketch the shapes of a 2s and a $2p_x$ orbital on the axes in Fig. 1.1.

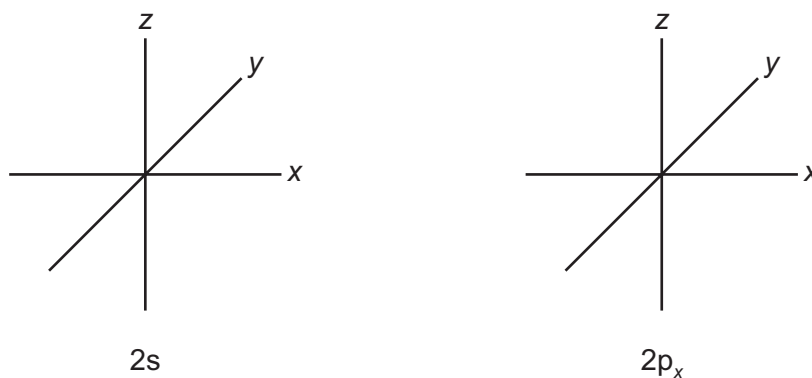


Fig. 1.1

[2]

50. O/N 22/P22/Q2(c)(i)

Context

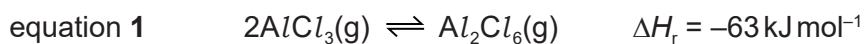
2 The chlorides of some of the Period 3 elements are shown in Table 2.1.

Table 2.1

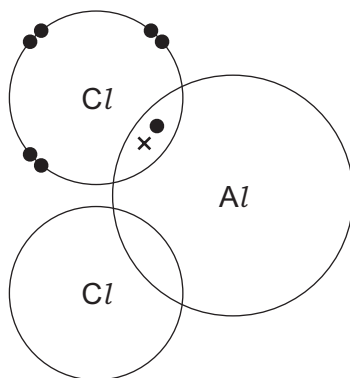
Period 3 chloride	NaCl	AlCl ₃	SiCl ₄	PCl ₅	PCl ₃	SCl ₂
bonding					C	C
structure					S	S
oxidation state of Period 3 element						

Context: part (c) (introduction)

(c) In the gas phase, AlCl₃(g) exists at equilibrium with Al₂Cl₆(g) as shown.



(i) Complete the dot-and-cross diagram to show the bonding in Al₂Cl₆.



[2]

51. O/N 22/P22/Q2(e)(ii)

Context

2 The chlorides of some of the Period 3 elements are shown in Table 2.1.

Table 2.1

Period 3 chloride	NaCl	AlCl ₃	SiCl ₄	PCl ₅	PCl ₃	SCl ₂
bonding					C	C
structure					S	S
oxidation state of Period 3 element						

(ii) SCl₂ has $M_r = 103.1$ and is a liquid at room temperature. SBr₂ has $M_r = 191.9$ and is a gas at room temperature.

Explain the difference in the physical state of SCl₂ and SBr₂. Give your answer in terms of intermolecular forces.

[2]

52. O/N 22/P22/Q2(f)(i)

Context

- 2** The chlorides of some of the Period 3 elements are shown in Table 2.1.

Table 2.1

Period 3 chloride	NaCl	AlCl ₃	SiCl ₄	PCl ₅	PCl ₃	SCl ₂
bonding					C	C
structure					S	S
oxidation state of Period 3 element						

Context: part (f) (introduction)

- (f)** Bismuth is a dense metal in the same group as phosphorus.

- (i)** Draw a labelled diagram to show the bonding in bismuth metal.

[2]

53. O/N 22/P23/Q2(a)

Context

- 2** Magnesium shows reactions typical of a Group 2 metal.

- (a)** Draw a labelled diagram to show the bonding in magnesium metal.

[2]

54. O/N 22/P23/Q3(d)

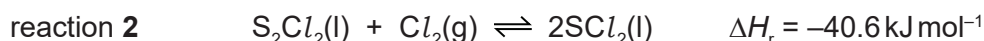
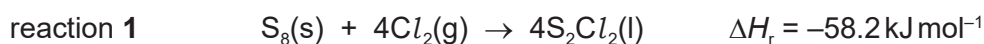
also Unit 4 (2 marks), Unit 5 (2 marks), Unit 13 (2 marks), Unit 7 (1 mark); this unit 4 marks

Context

- 3** Some of the common chlorides of Period 3 elements are shown in the list.



- (d)** Sulfur, S₈, reacts with chlorine to form several different chlorides. The most common are S₂Cl₂ and SCl₂. SCl₂ forms when sulfur reacts with an excess of chlorine.



- (i)** SCl₂ is a cherry-red liquid that reacts vigorously with water to form an acidic solution.

Use this information to deduce the bonding and structure shown by SCl₂.
Explain your answer.

[2]

- (ii)** Calculate the enthalpy change of formation, ΔH_f , of SCl₂(l). You may find it useful to use Hess's Law to construct an energy cycle.

enthalpy change of formation of SCl₂(l), $\Delta H_f =$ kJ mol⁻¹
[2]

- (iii) State the effect of a decrease in pressure on the position of equilibrium in reaction 2. Explain your answer.

[1]

Fig. 3.1 shows the two structural isomers of S_2Cl_2 .

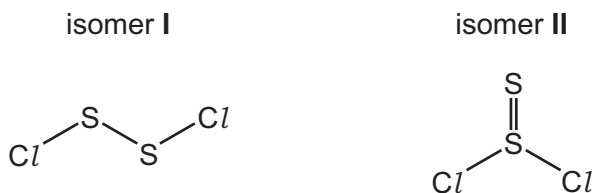


Fig. 3.1

- (iv) Define the term structural isomer.

[2]

- (v) Suggest a value for the $Cl-S-S$ bond angle in isomer I. Explain your answer.

bond angle = °

explanation

[2]

- (vi) Draw a dot-and-cross diagram to show the bonding in isomer II. Show outer shell electrons only.

[2]

55. M/J 22/P21/Q2(a)

Context

- 2 Nitrogen molecules, $N_2(g)$, contain two atoms attracted to each other by a triple covalent bond.

- (a) Describe how the triple covalent bond forms in a $N_2(g)$ molecule. Refer to orbital overlap and hybridisation in your answer.

[3]

56. M/J 22/P22/Q2(b)

Context

- 2 Radium, Ra, is an element found in Group 2 of the Periodic Table. It is a crystalline solid at room temperature and conducts electricity.

Radium chloride, $RaCl_2$, has a melting point of $900^\circ C$ and is soluble in water.

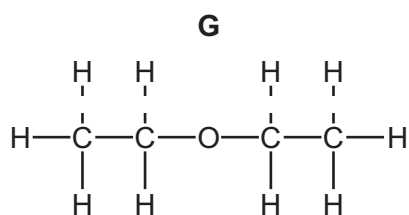
- (b) Draw a dot-and-cross diagram to show the arrangement of outer electrons in $RaCl_2$.

[1]

57. M/J 22/P23/Q3(a)(i)

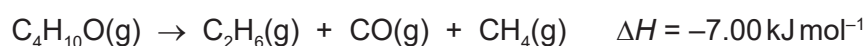
Context

3 **G** belongs to a group of compounds called ethers.

**Fig. 3.1**

Context: part (a) (introduction)

(a) When **G**, $\text{C}_4\text{H}_{10}\text{O}$, is heated, thermal decomposition occurs.



- (i)** The atoms in a molecule of CO are held together by a triple covalent bond. One of these bonds is a coordinate (dative covalent) bond.

Draw a dot-and-cross diagram to show the arrangement of outer electrons in a CO molecule.

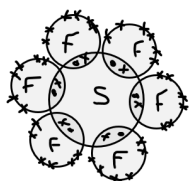
Use $\bullet$ to represent electrons from an oxygen atom.

Use $\times$ to represent electrons from a carbon atom.

[1]

Answers Section

1. M/J 26/P21/Q1(a)

1(a)(i)		1	A any symbol for electrons – symbols can all be the same or different A electrons can be on the lines or in the spaces A non-bonding electrons need not drawn in pairs I inner electrons
1(a)(ii)	octahedral	1	A square bipyramidal A octahedron A near miss in spelling e.g. octohedral I octagonal / octagon

2. M/J 26/P21/Q1(c)

1(c)(i)	M1 104.5(°) M2 2 bonding pairs (of electrons) and 2 lone pairs (of electrons) M3 lone pairs (of electrons) repel more than bonding pairs (of electrons)	3	M1 A range 103–105° inclusive I reference to stated shape even if incorrect I reference to tetrahedral geometry and 109.5 in text A 'bonding pairs' and 'lone pairs' assuming they are describing electrons unless stated otherwise. A 2 shared pairs for 2 bonding pairs For 2 lone pairs: A 2 pair of lone pairs A 2 non-bonding pairs I S—Cl bond / 2 bonds for bonding pair I any electrons on outer shell of chlorine R one unpaired electron If M2 is not awarded from text, award M2 for a suitably annotated Figure 1.2 or dot-and-cross diagram The electrons in both bond pairs and lone pairs must be shown on Figure 1.2 M3 A repulsion from lone pair greater than from bond pair M3 A repulsion lone pair – lone pair > lone pair – bonding pair > bonding pair – bonding pair M3 A repulsion lone pair – bonding pair > bonding pair – bonding pair Max 1 for 2 bonding pairs and 1 lone pair AND between 107–109° inclusive
1(c)(ii)	S as δ+ AND Both Cl as δ-	1	A δ+ δ+ / δ2+ / 2+δ on S A +δ and -δ I Δ I any electrons drawn on the atoms R ± without δ BOD TWO overall dipole direction arrows parallel to bonds I ONE overall dipole direction arrow / arrow

3. M/J 26/P21/Q1(d)

1(d)(i)		SCl ₂	H ₂ O	1	
	total number of electrons per molecule	50	10		

1(d)(ii)	SCl_2 AND greater number of electrons (per molecule) OR H_2O has weaker id (id) as it has smaller number of electrons (per molecule)	1 greater = higher = larger = bigger = more M1 awarded if incorrect number of electrons are given in 1(d)(i) AND still greater than 10 A ecf from 1(d)(i) for $\text{SCl}_2 < \text{H}_2\text{O}$ AND fewer (than 10) electrons per molecule if less than 10 shown in 1(d)(i) A ecf from 1(d)(i) $\text{SCl}_2 = \text{H}_2\text{O}$ AND same number of electrons per molecule if 10 shown in 1(d)(i) I greater electron density / electron cloud I SCl_2 has more valence / outer electrons
1(d)(iii)	electronegativity difference (between H and O) is greater (than S and Cl)	1 A larger for greater / bigger owtte A H–O has large electronegativity difference and S–Cl has small electronegativity difference I H_2O is more polar than SCl_2 I oxygen/O is more electronegative than chlorine I any comments on hydrogen bonding R H_2O is more electronegative than SCl_2 R O₂ is more electronegative than Cl₂

4. M/J 26/P22/Q1(b)

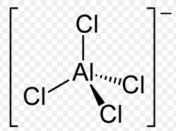
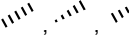
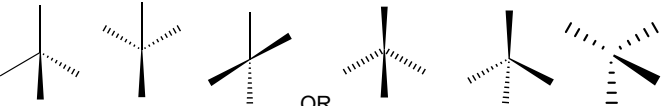
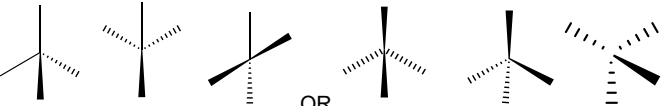
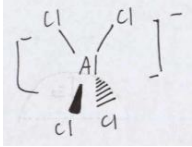
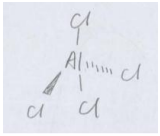
1(b)		2 M1 correct charges M1 A –2 for O ion and +1 / 1+ for Li ion M2 all else fully correct I symbols for Li and O A any symbol for electrons Max 1 for incorrect identification of type of bonding as covalent as long as all else correct Max 1 for any ionic compound with a positively charged species and a negatively charged species
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5. M/J 26/P24/Q1(c)

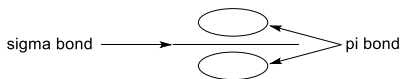
1(c)	Lone pair (of electrons) on a Cl (could be seen as : on a diagram) AND donated / shared with Al (could be seen by the arrow on a diagram) OR A fully annotated Figure 1.2:	1 A full annotations on Figure 1.2 i.e. lone pair AND dative arrow on <u>both</u> Cl to Al bonds R dative bond shown from Al to Cl on diagram R chloride / chlorine ion BOD lone pair A chlorine A donates / gives / shares / provides a lone pair / electron pair / both electrons <u>to Al</u>
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6. M/J 26/P24/Q1(d)

1(d)(i)	M1 (name of shape) tetrahedral M2 (bond angle) 109–109.5°	2 M1 A tetrahedra M1 A phonetically spelt name of shape
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1(d)(ii)	3D shape required:  Acceptable 3D descriptions of different bond types behind the plane and out of the plane are shown below  are all interchangeable  are all interchangeable Although the 3D structure shown in the syllabus is drawn as the example shown in the mark scheme it is acceptable to allow a combination of bond types shown below (3D structure has two types of wedge adjacent)  OR  A BOD any structures which show unambiguous use of bond shapes representing bonds behind the plane and out of the plane as long as they are different.	1 I negative charge (not required) I bond angles I dative bonds A BOD  R 
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7. M/J 26/P24/Q2(a)(i)

2(a)(i)	M1 sideways overlap of (adjacent) p orbitals M2 above and below the (covalent/σ) bond	2 M1 A unambiguous labelled diagram A 'sideways overlap of p' A sideways = lateral = side on = side by side = side to side = parallel I indirect I horizontal / vertical I shoulder-to-shoulder M2 A above and below the (covalent) bond A labelled diagram: 
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8. MAR 26/P22/Q2(a)(ii)

2(a)(ii)		1
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9. MAR 26/P22/Q4(b)

4(b)	M1 clearly distinct / labelled H bond M2 from lone pair on any oxygen to H atom attached to O M3 with three correct successive partial charges across two molecules	3
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10. O/N 25/P22/Q3(d)(ii)

3(d)(ii)	M1 B < A < hexane. M2 hexane / unbranched has stronger / more IMFs M3 hexane / unbranched requires most / more energy	3
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11. O/N 25/P24/Q1(b)(v)

1(b)(v)	M1 (trigonal) pyramidal M2 three bonding pairs AND one lone pair (of electrons)	2
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12. M/J 25/P21/Q1(a)(ii)

1(a)(ii)	delocalised electrons move (through the structure)	1
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13. M/J 25/P23/Q4(b)

4(b)(i)	M1 non-linear M2 bond angle – $120 > x \geq 116^\circ$	2
4(b)(ii)	M1 weaker dipole moment (in SO_2) M2 electronegativity (difference) is larger (/ 1.4) for H and O than for S and O (/ 0.9) OR S=O bond is less polar than H—O OR stronger dipoles in S=O / SO_2	2

14. M/J 25/P24/Q2(a)

2(a)(i)	non-linear	1
2(a)(ii)	104.5°	1

15. M/J 25/P24/Q3(d)(i)

3(d)(i)	(electrostatic) attraction between the nuclei (of two atoms) and a shared pair of electrons	1
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16. MAR 25/P22/Q1(b)(ii)

1(b)(ii)	trigonal bipyramid(al)	1
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17. MAR 25/P22/Q2(a)

2(a)	electrostatic attraction between positive metal ions / cations and delocalised electrons	2
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18. O/N 24/P21/Q3(a)(iii)

3(a)(iii)	its delocalised electrons are free to move	1
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19. O/N 24/P21/Q3(d)

3(d)(i)	M1 molecules CH_4 to PbH_4 have greater number of electrons / number of electrons increase	1
	M2 greater / stronger instantaneous dipole–induced forces / London forces / dispersion forces OR more energy required to overcome the instantaneous dipole–induced forces / London forces / dispersion forces	1
3(d)(ii)	tetrahedral	1

20. O/N 24/P22/Q2(a)(ii)

2(a)(ii)	Al^{3+} 	1
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21. O/N 24/P22/Q2(c)

2(c)(i)	(molecules) H_2S to H_2Te / they have greater number of electrons	1
	stronger instantaneous dipole–induced dipole / London dispersion forces OR more energy required to overcome the instantaneous dipole–induced dipole / London dispersion forces	1
2(c)(ii)	(only) H_2O has hydrogen bonding AND hydrogen bonding (much) stronger than the other intermolecular forces OR H_2O has hydrogen bonding AND because of higher electronegativity of O compared to S OR stronger van der Waals' forces because (only) H_2O has hydrogen bonding OR hydrogen bond AND increased strength of permanent dipoles in H_2O outweighs the increase in strength of id-id in the others	1

22. O/N 24/P22/Q3(b)(iv)

3(b)(iv)	tetrahedral	1
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23. O/N 24/P22/Q3(f)(i)

3(f)(i)	M1 one sigma / σ bond AND head-on (overlap of) p / sp (orbitals)	1
	M2 two pi / π bond(s) AND side-on (overlap of / involving) p (orbitals)	1

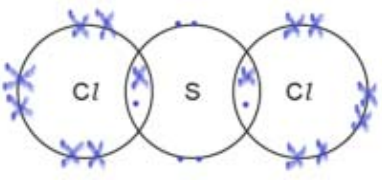
24. O/N 24/P23/Q3(a)(iii)

3(a)(iii)	its delocalised electrons are free to move	1
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25. O/N 24/P23/Q3(d)

3(d)(i)	M1 molecules CH_4 to PbH_4 have greater number of electrons / number of electrons increase	1
	M2 greater / stronger instantaneous dipole–induced forces / London forces / dispersion forces OR more energy required to overcome the instantaneous dipole–induced forces / London forces / dispersion forces	1
3(d)(ii)	tetrahedral	1

26. M/J 24/P21/Q2(b)

2(b)(i)	M1 1 shared pair of electrons between each S–Cl shown as $\bullet\bullet$ M2 6 non-bonding electrons around each Cl shown as $\times$ AND 4 non-bonding electrons around S shown as $\bullet$	2
		
2(b)(ii)	M1 shape non-linear M2 bond angle $103\text{--}105^\circ$	2

27. M/J 24/P21/Q5(a)

5(a)	(electrostatic) attraction between nuclei of two atoms and shared pair of electrons	1
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28. M/J 24/P22/Q1(c)

1(c)	positively charged ions delocalised electrons labelled diagram shows: - regular arrangement of minimum of 2×2 circles containing + for positive ions AND surrounded by (sea of) delocalised electrons	1
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29. M/J 24/P22/Q3(d)

3(d)	M1 6 electrons between C and N AND 2 electrons between C and S M2 3 lone pairs around S AND none around C AND 1 lone pair around N	2
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30. M/J 24/P23/Q1(a)(ii)

1(a)(ii)	<table border="1"> <thead> <tr> <th></th><th>σ bond</th><th>π bond</th></tr> </thead> <tbody> <tr> <td>no of bonds</td><td>1</td><td>2</td></tr> <tr> <td>how orbitals overlap</td><td>direct (orbital overlap)</td><td>sideways (overlap of adjacent p orbitals)</td></tr> </tbody> </table>		σ bond	π bond	no of bonds	1	2	how orbitals overlap	direct (orbital overlap)	sideways (overlap of adjacent p orbitals)	4
	σ bond	π bond									
no of bonds	1	2									
how orbitals overlap	direct (orbital overlap)	sideways (overlap of adjacent p orbitals)									

31. M/J 24/P23/Q4(a)

4(a)	OR Correct representation or label showing M1 (ionic bond between) NH_4^+ and Cl^- AND 4 N—H covalent bonds OR 4 shared pairs of electrons in NH_4 M2 1 N—H coordinate bond OR arrow on bond/line in correct direction or only 1 bonding pair using the same symbol for electrons	2
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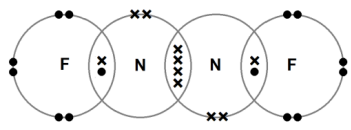
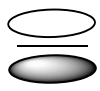
32. MAR 24/P22/Q1(a)

1(a)	labelled diagram shows: - regular arrangement of positive ions AND surrounded by / (sea of) delocalised electrons	1
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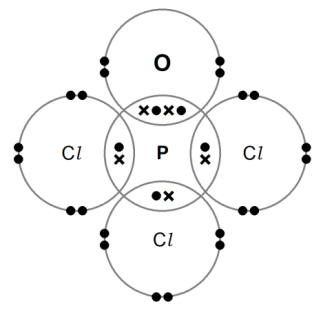
33. MAR 24/P22/Q2(c)(i)

2(c)(i)	• trigonal bipyramidal • octahedral	2
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34. MAR 24/P22/Q3(d)

3(d)(i)	 correct bonding pairs	1
	lone pairs	1
3(d)(ii)	sp^2	1
3(d)(iii)	 M1 shows the pi (π) orbital	1
	M2 refers to overlap of (two) (2)p orbitals side-on / above and below the plane	1

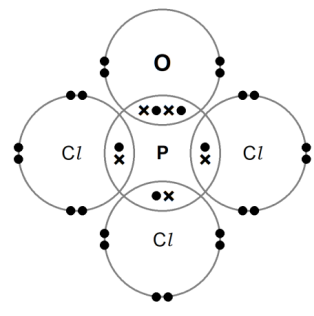
35. O/N 23/P21/Q1(d)(iii)

1(d)(iii)	 Correct bonding e^-	1
	rest of electrons correct (i.e. to make 32 electrons in total)	1

36. O/N 23/P22/Q4(a)(iii)

4(a)(iii)	dashed (or distinctly labelled) bond between O and H in different molecules of H-O-H and propanoic acid	1
	correct sequence of three partial charges over the H-bond $H^{\delta+} \quad O^{\delta-} \cdots H^{\delta+}$ OR $O^{\delta-} \quad H^{\delta+} \cdots O^{\delta-}$	1
	lone pair on O in line with H-bond	1

37. O/N 23/P23/Q1(d)(iii)

1(d)(iii)	 Correct bonding e^-	1
	rest of electrons correct (i.e. to make 32 electrons in total)	1

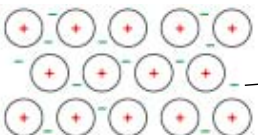
38. M/J 23/P21/Q1(e)

1(e)	M1 6 bonding pairs (of electrons and 0 lone pairs) OR coordination number = 6 OR x = 6 M2 bonding pairs repel (equally)	2
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39. M/J 23/P22/Q1(b)

1(b)(i)	power of an atom to attract electrons to itself	1
1(b)(ii)	(across a period) - increase in nuclear charge - similar shielding (so) increase in nuclear attraction for bonding / outer / valence electrons OR bonding / outer / valence electron(s) are more strongly attracted to nucleus Two correct for one mark, three correct for two marks	2
1(b)(iii)	hydrogen bond	1
1(b)(iv)	M1 link shown as a dashed line between the lone pair of electrons from N of one NH ₃ to one H on other NH ₃ M2 minimum 3 correct partial charges (on adjacent atoms) over two NH ₃ molecules EITHER $\delta^- \text{N} - \delta^+ \text{H} \cdots \delta^- \text{N}$ OR $\delta^+ \text{H} \cdots \delta^- \text{N} - \delta^+ \text{H}$	2
1(b)(v)	M1 O is more electronegative than N M2 two H-bonds per water molecule : 1 per ammonia molecule.	2

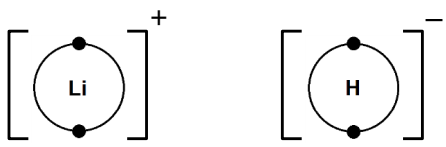
40. M/J 23/P23/Q1(a)(ii)

1(a)(ii)	 delocalised electrons M1 diagram showing minimum of 4 particles - circles containing just a + do not have to be labelled - empty circles / circles with Cu must be labelled + ion / positive ion / cation / Cuⁿ⁺ AND with the circles surrounded by electrons shown as e⁻ / – OR little circles labelled delocalised electrons	1
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41. MAR 23/P22/Q1(a)

1(a)(i)	power of an atom to attract electrons to itself	1
1(a)(ii)	- O lower nuclear charge / lower proton number - O has (one) fewer shell than S / less shielding - greater attraction (for nucleus) in O	2

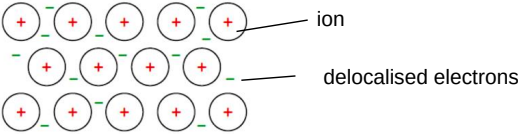
42. MAR 23/P22/Q1(b)

1(b)(i)		2
1(b)(ii)	non-linear	1

43. MAR 23/P22/Q1(e)(i)

1(e)(i)						2
	compound	O=C=O	O=S=O	S=C=S	S=C=O	
	overall dipole moment		✓		✓	

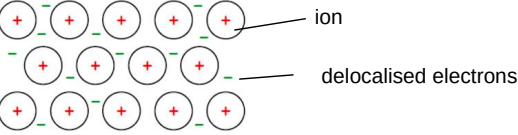
44. MAR 23/P22/Q2(a)

2(a)(i)	 M1 diagram showing minimum of 4 particles (in total in 2 rows) - circles containing X^{n+} do not have to be labelled - must be labelled as 'ion' OR empty circles / circles with X must be labelled + ion / positive ion / cation / X^{n+} AND with the circles surrounded by electrons shown as e^- / – OR little circles labelled electrons M2 label / legend showing delocalised electrons	1
2(a)(ii)	Mg has more delocalised e^- (than Na)	1

45. MAR 23/P22/Q3(d)(iv)

3(d)(iv)	instantaneous dipole—induced dipole / id—id AND permanent dipole—permanent dipole / pd—pd	1
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46. O/N 22/P21/Q2(a)

2(a)	 M1 diagram showing minimum of 4 particles (in total in two rows) (circles) - circles containing Mg^{n+} do not have to be labelled - must be labelled as 'ion' OR empty circles / circles with Mg must be labelled + ion / positive ion / cation / Mgⁿ⁺ AND - circles surrounded by electrons shown as e^- / – OR in an area around the circles labelled as 'electrons' OR little circles labelled electrons OR electrons drawn only on perimeter of structure M2 label / legend showing <u>delocalised</u> electrons	1
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47. O/N 22/P21/Q3(d)

3(d)(i)	M1 (structure $\Rightarrow$) simple / molecular, because it has a low melting / boiling point	1
	M2 (bonding $\Rightarrow$) covalent, because it is hydrolysed	1
3(d)(ii)	Correct use of enthalpy values / correct direction / use of arrows (with or without an energy cycle) M1 $((-58.2) + (-40.6))$ OR (-98.8) M2 $[(-58.2 + 4(-40.6) = 8\Delta H_f)]$ Correct calculation and correct stoichiometry $\Delta H_f = -27.6 \text{ (kJ mol}^{-1}\text{)}$	2
3(d)(iii)	equilibrium moves to left AND more moles / molecules of gas on LHS	1
3(d)(iv)	(molecules / isomers with) the same molecular formula	1
	but different structural formulae	1
3(d)(v)	M1 $90^\circ < Cl-S-S \leq 108^\circ$	1
	M2 sulfur has two lone pairs (of e^-) (and two bonding pairs) AND repulsion from lone pairs (greater)	1

3(d)(vi)	bonding electrons	1
	all other electrons correct	1

48. O/N 22/P22/Q1(a)

1(a)	M1 ionic = oppositely charged ions / cations AND anions	1
	M2 covalent = shared pair(s) of electrons AND two nuclei	1

49. O/N 22/P22/Q1(c)(iii)

1(c)(iii)	2s 2p_x	2
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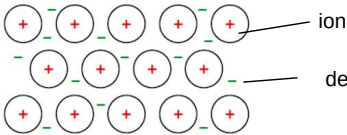
50. O/N 22/P22/Q2(c)(i)

2(c)(i)		1
	<i>'dot-and-cross' diagram to show</i> M1 correct number and connectivity of Cl and Al atoms M2 2 correct dative bonds (••) between Cl and Al of AlCl₃ neighbour AND correct number of non-bonding electrons around each atom of Al₂Cl₆	1

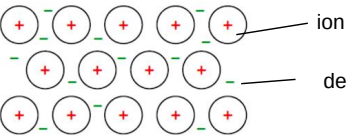
51. O/N 22/P22/Q2(e)(ii)

2(e)(ii)	M1 sum of instantaneous dipole – dipole attractions and permanent dipole attractions in SCl₂ is greater than those in SBr₂ OR permanent dipole-permanent dipole (forces) in SCl₂ are stronger OR intermolecular forces are stronger in SCl₂ ora	1
	M2 SCl₂ (molecule / bond) is more polar than SBr₂ OR Cl is more electronegative than Br so SCl₂ has stronger (permanent) dipoles ora	1

52. O/N 22/P22/Q2(f)(i)

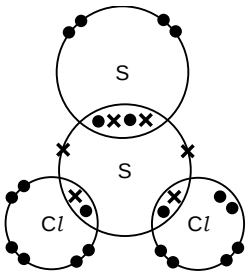
2(f)(i)	 M1 diagram showing minimum of 4 particles (in total in 2 rows) - circles containing Bi^{n+} do not have to be labelled - must be labelled as 'ion' OR empty circles / circles with Bi must be labelled + ion / positive ion / cation / Bi^{n+} AND - with the circles surrounded by electrons shown as e^- / – OR little circles labelled electrons	1
	M2 label / legend showing <u>delocalised</u> electrons	1

53. O/N 22/P23/Q2(a)

2(a)	 M1 diagram showing minimum of 4 particles (in total in two rows) (circles) - circles containing Mg^{n+} do not have to be labelled - must be labelled as 'ion' OR empty circles / circles with Mg must be labelled + ion / positive ion / cation / Mg^{n+} AND - circles surrounded by electrons shown as e^- / – OR in an area around the circles labelled as 'electrons' OR little circles labelled electrons OR electrons drawn only on perimeter of structure	1
	M2 label / legend showing <u>delocalised</u> electrons	1

54. O/N 22/P23/Q3(d)

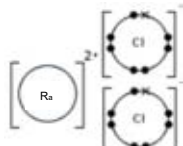
3(d)(i)	M1 (structure =) simple / molecular, because it has a low melting / boiling point	1
	M2 (bonding =) covalent, because it is hydrolysed	1
3(d)(ii)	Correct use of enthalpy values / correct direction / use of arrows (with or without an energy cycle) M1 $(-58.2) + (-40.6)$ OR (-98.8) M2 $[(-58.2 + 4(-40.6) = 8\Delta H_f)]$ Correct calculation and correct stoichiometry $\Delta H_f = -27.6 \text{ (kJ mol}^{-1}\text{)}$	2
3(d)(iii)	equilibrium moves to left AND more moles / molecules of gas on LHS	1
3(d)(iv)	(molecules / isomers with) the same molecular formula	1
	but different structural formulae	1
3(d)(v)	M1 $90^\circ < \text{Cl-S-S} \leq 108^\circ$	1
	M2 sulfur has two lone pairs (of e^-) (and two bonding pairs) AND repulsion from lone pairs (greater)	1

3(d)(vi)	 bonding electrons	1
	all other electrons correct	1

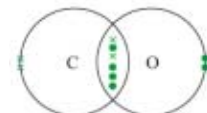
55. M/J 22/P21/Q2(a)

2(a)	M1 one sigma / σ bond and two pi / π bonds M2 sp hybridisation (in each N atom) M3 sigma / σ forms from direct / head-on / end-on overlap of orbitals AND pi / π forms sideways / lateral overlap of (p) orbitals	3
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56. M/J 22/P22/Q2(b)

2(b)	Ra^{2+} and 2 x Cl^- 0 electrons surrounding 2Ra AND 8 electrons surrounding Cl 	1
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57. M/J 22/P23/Q3(a)(i)

3(a)(i)		1
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